

*Note: Product Name, Item Code and Pharma code Position & Orientation will be changed based on folding feasibility

Folding Pattern 32 mm

Folding Pattern 56 mm

1313001030-01

ESOME

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ESOME

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ESOME

ESOME Esomeprazole Sodium Powder For Solution For Injection or Infusion (Lyophilized powder)
40mg/Vial
COMPOSITION:
Each vial contains:-
Esomeprazole Sodium equivalent to esomeprazole.....40 mg
Disodium Edetate USP 1.5 mg
Sodium Hydroxide USP-NF q.s
Water for Injection BP q.s to 1.5 mL
DOSAGE FORM
Powder for solution for Injection or Infusion (Lyophilized powder)
PRODUCT DESCRIPTION
Esomeprazole Sodium for Injection is a lyophilized cake/powder for Injection containing 42.5 mg of Esomeprazole Sodium equivalent to 40 mg of Esomeprazole. The product is packed as 40 mg of Esomeprazole in 5 mL USP Type I clear glass vial, stoppered with 13 mm Grey Bromobutyl Lyo rubber stoppers and sealed with 13 mm aluminium flip-off seals with Royal blue colour button.
DESCRIPTION AFTER RECONSTITUTION
Product is clear colorless to slight yellow color solution after reconstitution.
PHARMACODYNAMICS
Pharmacotherapeutic group: Proton pump inhibitor; ATC code: A02BC05
Esomeprazole is the S-isomer of omeprazole and reduces gastric acid secretion through a specific targeted mechanism of action. It is a specific inhibitor of the acid pump in the parietal cell. Both the R- and S-isomer of omeprazole have similar pharmacodynamic activity.
Site and mechanism of action
Esomeprazole is a weak base and is concentrated and converted to the active form in the highly acidic environment of the secretory canaliculi of the parietal cell, where it inhibits the enzyme H+K+-ATPase – the acid pump and inhibits both basal and stimulated acid secretion.
Effect on gastric acid secretion
After 5 days of oral dosing with 20 mg and 40 mg of esomeprazole, intragastric pH above 4 was maintained for a mean time of 13 hours and 17 hours, respectively over 24 hours in symptomatic GERD patients. The effect is similar irrespective of whether esomeprazole is administered orally or intravenously.
Using AUC as a surrogate parameter for plasma concentration, a relationship between inhibition of acid secretion and exposure has been shown after oral administration of esomeprazole.
During intravenous administration of 80 mg esomeprazole as a bolus infusion over 30 minutes followed by a continuous intravenous infusion of 8 mg/h for 23.5 hours, intragastric pH above 4, and pH above 6 was maintained for a mean time of 21 hours, and 11-13 hours, respectively, over 24 hours.
Therapeutic effects of acid inhibition
Healing of reflux esophagitis with esomeprazole 40 mg occurs in approximately 78% of patients after 4 weeks, and in 93% after 8 weeks of oral treatment.
Other effects related to acid inhibition
During treatment with antisecretory medicinal products serum gastrin increases in response to the decreased acid secretion. Also CgA increases due to decreased gastric acidity. The increased CgA level may interfere with investigations for neuroendocrine tumours. Available published evidence suggests that proton pump inhibitors should be discontinued between 5 days and 2 weeks prior to CgA measurements. This is to allow CgA levels that might be spuriously elevated following PPI treatment to return to reference range.
An increased number of ECL cells possibly related to the increased serum gastrin levels. During long-term oral treatment with antisecretory drugs gastric glandular cysts have been reported to occur at a somewhat increased frequency. These changes are a physiological consequence of pronounced inhibition of acid secretion, are benign and appear to be reversible. Decreased gastric acidity due to any means including proton pump inhibitors, increases gastric counts of bacteria normally present in the gastrointestinal tract. Treatment with proton pump inhibitors may lead to slightly increased risk of gastrointestinal infections such as *Salmonella* and *Campylobacter* and, in hospitalised patients, possibly also *Clostridium difficile*.
Mechanism of Action
Esomeprazole is the S-isomer of omeprazole and reduces gastric acid secretion through a specific targeted mechanism of action. It is a specific inhibitor of the acid pump in the parietal cell. Both the R- and S-isomer of omeprazole have similar pharmacodynamic activity.
Pharmacokinetics
Distribution
The apparent volume of distribution at steady state is approximately 0.22 l/kg body weight. Esomeprazole is 97% plasma protein bound.
Metabolism and excretion
Esomeprazole is completely metabolised by the cytochrome P450 system (CYP). The major part of the metabolism of esomeprazole is dependent on the polymorphic CYP2C19, responsible for the formation of the hydroxy- and desmethyl metabolites of esomeprazole. The remaining part is dependent on another specific isoform, CYP3A4, responsible for the formation of esomeprazole sulphone, the main metabolite in plasma. The parameters below reflect mainly the pharmacokinetics in individuals with a functional CYP2C19 enzyme, extensive metabolisers. Total plasma clearance is about 17 l/h after a single dose and about 9 l/h after repeated administration. The plasma elimination half-life is about 1.3 hours after repeated once-daily dosing. Total exposure (AUC) increases with repeated administration of esomeprazole. This increase is dose-dependent and results in a non-linear dose-AUC relationship after repeated administration. This time-and dose- dependency is due to a decrease of first pass metabolism and systemic clearance probably caused by inhibition of the CYP2C19 enzyme by esomeprazole and/or its sulphone metabolite.
Esomeprazole is completely eliminated from plasma between doses with no tendency for accumulation during once-daily administration. Following repeated doses of 40 mg administered as intravenous injections, the mean peak plasma concentration is approx. 13.6 micromol/l. The mean peak plasma concentration after corresponding oral doses is approx. 4.6 micromol/l. A smaller increase (of approx. 30%) can be seen in the total exposure after intravenous administration compared to oral administration. There is a dose-linear increase in total exposure following intravenous administration of esomeprazole as a 30-minute infusion (40 mg, 80 mg or 120 mg) followed by a continuous infusion (4 mg/h or 8 mg/h) over 23.5 hours. The major metabolites of esomeprazole have no effect on gastric acid secretion. Almost 80% of an oral dose of esomeprazole is excreted as metabolites in the urine, the remainder in the faeces. Less than 1% of the parent drug is found in urine. The metabolism of esomeprazole is not significantly changed in elderly (71-80 years of age). The metabolism of esomeprazole in patients with mild to moderate liver dysfunction may be impaired. The metabolic rate is decreased in patients with severe liver dysfunction resulting in a doubling of the total exposure of esomeprazole. Therefore, a maximum dose of 20 mg should not be exceeded in GERD patients with severe dysfunction. For patients with bleeding ulcers and severe liver impairment, following an initial bolus dose of 80 mg, a maximum continuous intravenous infusion dose of 4 mg/h for 71.5 hours may be sufficient. Esomeprazole or its major metabolites do not show any tendency to accumulate with once daily dosing. Since the kidney is responsible for the excretion of the metabolites of esomeprazole but not for the elimination of the parent compound, the metabolism of esomeprazole is not expected to be changed in patients with impaired renal function. No studies have been performed in patients with decreased renal function.
INDICATION
ESOME for injection and infusion is indicated for

- Gastric antisecretory treatment when the oral route is not possible, such as:
 - gastroesophageal reflux disease in patients with esophagitis and/or severe symptoms of reflux.
 - Healing of gastric ulcers associate with NSAID therapy.
 - Prevention of gastric and duodenal ulcers associated with NSAID therapy, in patients at risk.
- Prevention of rebleeding following therapeutic endoscopy for acute bleeding gastric or duodenal ulcers.

RECOMMENDED DOSE
Gastric antisecretory treatment when the oral route is not possible

Patients who cannot take oral medication may be treated parenterally with 20-40 mg once daily. Patients with reflux oesophagitis should be treated with 40 mg once daily. Patients treated symptomatically for reflux disease should be treated with 20 mg once daily. For healing of gastric ulcers associated with NSAID therapy the usual dose is 20 mg once daily. For prevention of gastric and duodenal ulcers associated with NSAID therapy, patients at risk should be treated with 20 mg once daily. Usually the intravenous treatment duration is short and transfer to oral treatment should be made as soon as possible. Treatment with ESOME IV can be given for up to 10 days as part of a full treatment period for the specified indications.

Prevention of rebleeding of gastric and duodenal ulcers

Following therapeutic endoscopy for acute bleeding gastric or duodenal ulcers, 80 mg should be administered as a bolus infusion over 30 minutes, followed by a continuous intravenous infusion of 8 mg/h given over 3 days (72 hours).

The parenteral treatment period should be followed by oral acid suppression therapy.

Method of Administration

For preparation of reconstituted solution, see "Instructions for Use and Handling". Injection

40 mg dose

5 ml of the reconstituted solution (8 mg/ml) should be given as an intravenous injection over a period of at least 3 minutes.

20 mg dose

2.5 ml or half of the reconstituted solution (8 mg/ml) should be given as an intravenous injection over a period of at least 3 minutes. Any unused solution should be discarded.

40 mg dose

The reconstituted solution should be given as an intravenous infusion over a period of 10 to 30 minutes.

20 mg dose

Half of the reconstituted solution should be given as an intravenous infusion over a period of 10 to 30 minutes. Any unused solution should be discarded.

80 mg bolus dose

The reconstituted solution should be given as a continuous intravenous infusion over 30 minutes.

8 mg/h dose

The reconstituted solution should be given as a continuous intravenous infusion over a period of 71.5 hours (calculated rate of infusion of 8 mg/h. See "Shelf Life" for shelf-life of the reconstituted solution).

Children and adolescents

ESOME should not be used in children since no data is available.

Impaired renal function

Dose adjustment is not required in patients with impaired renal function. Due to limited experience in patients with severe renal insufficiency, such patients should be treated with caution.

Impaired hepatic function

GERD: Dose adjustment is not required in patients with mild to moderate liver impairment. For patients with severe liver impairment, a maximum daily dose of 20 mg ESOME should not be exceeded.

Bleeding ulcers: Dose adjustment is not required in patients with mild to moderate liver impairment. For patients with severe liver impairment, following an initial bolus dose of 80 mg ESOME for infusion, a continuous intravenous infusion dose of 4 mg/h for 71.5 hours may be sufficient.

Elderly

Dose adjustment is not required in the elderly.

ROUTE OF ADMINISTRATION

To be administered intravenously

CONTRAINDICATION

Hypersensitivity to the active substance esomeprazole or to other substituted benzimidazoles or to any of the excipients of this medicinal product. Esomeprazole should not be used concomitantly with nelfinavir.

WARNING AND PRECAUTIONS

In the presence of any alarm symptom (e.g. significant unintentional weight loss, recurrent vomiting, dysphagia, haematemesis or melaena) and when gastric ulcer is suspected or present, malignancy should be excluded, as treatment with ESOME may alleviate symptoms and delay diagnosis. Treatment with proton pump inhibitors may lead to slightly increased risk of gastrointestinal infections such as *Salmonella* and *Campylobacter*. Co-administration of esomeprazole with atazanavir is not recommended. Esomeprazole, as all acid-blocking medicines, may reduce the absorption of Vitamin B12 (cyanocobalamin) due to hypo- or achlorhydria. This should be considered in patients with reduced body stores or risk factors for reduced Vitamin B12 absorption on long-term therapy. Esomeprazole is a CYP2C19 inhibitor. When starting or ending treatment with esomeprazole, the potential for interactions with drugs metabolised through CYP2C19 should be considered.

An interaction is observed between clopidogrel and esomeprazole. The relevance of this interaction is uncertain. As a precaution, concomitant use of esomeprazole and clopidogrel should be discouraged.

Regular Surveillance

Patients on proton pump inhibit treatment (particularly those treated for long term) should be kept under regular surveillance.

Subacute Cutaneous Lupus Erythematosus (SCLE)

Proton pump inhibitors are associated with very infrequent cases of subacute cutaneous lupus erythematosus (SCLE). If lesions occur, especially in sun-exposed areas of the skin, and if accompanied by arthralgia, the patient should seek medical help promptly and the health care professional should consider stopping esomeprazole. SCLE after previous treatment with a proton pump inhibitor may increase the risk of SCLE with other proton pump inhibitors.

Hypomagnesaemia

Severe hypomagnesaemia has been reported in patients treated with PPI like esomeprazole for at least three months, and in most cases for a year. Serious manifestations of hypomagnesaemia such as fatigue, tetany, delirium, convulsions, dizziness and ventricular arrhythmia can occur but they may begin insidiously and be overlooked. In most affected patients, hypomagnesaemia improved after magnesium replacement and discontinuation of the PPI. For patients expected to be on prolonged treatment or who take PPI with digoxin or drugs that may cause hypomagnesaemia (e.g. diuretics), health care professionals should consider measuring magnesium levels before starting PPI treatment and periodically during treatment.

Fracture

Proton pump inhibitors, especially if used in high doses and over long durations (>1 year), may modestly increase the risk of hip, wrist and spine fracture, predominantly in the elderly or in presence of other recognised risk factors. Patients at risk of osteoporosis should receive care according to current clinical guidelines and they should have an adequate intake of vitamin D and calcium.

Clostridium Difficile Diarrhea

Published observational studies suggest that PPI therapy may be associated with an increased risk of Clostridium difficile associated diarrhea, especially in hospitalized patients. This diagnosis should be considered for diarrhea that does not improve. Patients should use the lowest dose and shortest duration of PPI therapy appropriate to the condition being treated.

Vitamin B12 Deficiency

Daily treatment with any acid-suppressing medications over a long period of time (e.g., longer than 3 years) may lead to malabsorption of cyanocobalamin (vitamin B12) caused by hypo- or achlorhydria. Rare reports of cyanocobalamin deficiency occurring with acid-suppressing therapy have been reported in the literature. This diagnosis should be considered if clinical symptoms consistent with cyanocobalamin deficiency are observed.

Interference with laboratory tests

Increased Chromogranin A (CgA) level may interfere with investigations for neuroendocrine tumours. If the patient(s) are due to have a test on Chromogranin A level, [product name] treatment should be stopped for at least 5 days before CgA measurements to avoid this interference (see section Pharmacodynamic). If CgA and gastrin levels have not returned to reference range after initial measurement, measurements should be repeated 14 days after cessation of proton pump inhibitor treatment.

240 mm

320 mm

Pharma Code: XXX

INTERACTIONS WITH OTHER MEDICAMENTS

Interaction studies have only been performed in adults.

Effects of esomeprazole on the pharmacokinetics of other drugs

a) Medicinal products with pH dependent absorption

Gastric acid suppression during treatment with esomeprazole and other PPIs might decrease or increase the absorption of medicinal products drugs with a gastric pH dependent absorption. As with other medicinal products that decrease the intragastric acidity, the absorption of medicinal products, such as ketoconazole, itraconazole and erlotinib and the absorption of digoxin can increase during treatment with esomeprazole. Concomitant treatment with omeprazole (20 mg daily) and digoxin increased the bioavailability of digoxin by 10%. Digoxin toxicity has been rarely reported. However, caution should be exercised when esomeprazole is given at high doses in elderly patients. Therapeutic drug monitoring of digoxin should then be reinforced.

Omeprazole has been reported to interact with some protease inhibitors. Increased gastric pH during omeprazole treatment may change the absorption of the protease inhibitors. Other possible interaction mechanisms are via inhibition of CYP 2C19.

For atazanavir and nelfinavir, decreased serum levels have been reported when given together with omeprazole and concomitant administration is not recommended. Due to the similar pharmacodynamic effects and pharmacokinetic properties of omeprazole and esomeprazole, concomitant administration with esomeprazole and atazanavir is not recommended and concomitant administration with esomeprazole and nelfinavir is contraindicated.

b) Drugs metabolised by CYP2C19

Esomeprazole inhibits CYP2C19, the major esomeprazole metabolising enzyme. Thus, when esomeprazole is combined with drugs metabolised by CYP2C19, such as diazepam,citalopram, imipramine, clomipramine, phenytoin etc., the plasma concentrations of these drugs may be increased and a dose reduction could be needed. Concomitant oral administration of 30 mg esomeprazole resulted in a 45% decrease in clearance of the CYP2C19 substrate diazepam. Concomitant oral administration of 40 mg esomeprazole and phenytoin resulted in a 13% increase in trough plasma levels of phenytoin in epileptic patients. It is recommended to monitor the plasma concentrations of phenytoin when treatment with esomeprazole is introduced or withdrawn.

Omeprazole (40 mg once daily) increased voriconazole (a CYP2C19 substrate) Cmax and AUCτ by 15% and 41%, respectively. Monitoring is recommended when initiating and ending concomitant esomeprazole treatment during treatment with warfarin or other coumarine derivatives. Concomitant oral administration of 40 mg esomeprazole and cisapride resulted in a 32% increase in area under the plasma concentration-time curve (AUC) and a 31% prolongation of elimination half-life (t1/2) but no significant increase in peak plasma levels of cisapride. As a precaution concomitant use of clopidogrel should be discouraged.

c) Unknown mechanism

Concomitant administration of esomeprazole has been reported to increase the serum levels of tacrolimus. When given together with proton pump inhibitors, methotrexate levels have been reported to increase in some patients. In high-dose methotrexate administration a temporary withdrawal of esomeprazole may need to be considered.

d) Effects of other drugs on the pharmacokinetics of esomeprazole

Esomeprazole is metabolised by CYP2C19 and CYP3A4. Concomitant oral administration of esomeprazole and a CYP3A4 inhibitor, clarithromycin (500 mg b.i.d.), resulted in a doubling of the exposure (AUC) to esomeprazole. Concomitant administration of esomeprazole and a combined inhibitor of CYP2C19 and CYP 3A4 may result in more than doubling of the esomeprazole exposure. The CYP2C19 and CYP3A4 inhibitor voriconazole increased omeprazole AUCτ by 280%. However, dose adjustment should be considered in patients with severe hepatic impairment and if long-term treatment is indicated. Drugs known to induce CYP2C19 or CYP3A4 or both (such as rifampicin and St. John's wort) may lead to decreased esomeprazole serum levels by increasing the esomeprazole metabolism.

PREGNANCY AND LACTATION

For esomeprazole limited data on exposed pregnancies are available. Caution should be exercised when prescribing ESOME to pregnant women. It is not known whether esomeprazole is excreted in human breast milk. No studies in lactating women have been performed. Therefore ESOME should not be used during breastfeeding.

SIDE EFFECTS

The following adverse drug reactions have been identified or suspected for esomeprazole administered orally or intravenously. It is presented in Table 1: Presentation of Side Effects

Table 1 : Presentation of Side Effects

Category	Classification				
	Very rare	Rare	Common	Uncommon	Not known
Blood and lymphatic system disorders	Aggranulocytosis, pancytopenia	Leukopenia, thrombocytopenia			
Immune system disorders		Hypersensitivity reactions e.g. fever, angioedema and anaphylactic reaction/shock			
Metabolism and nutrition disorders	Hypomagnesaemia; severe hypomagnesaemia may result in hypocalcaemia, Hypomagnesaemia may also result in hypokalaemia.	Hyponatraemia		Peripheral oedema	
Psychiatric disorders	Aggression, hallucinations	Agitation, confusion, depression		Insomnia	
Nervous system disorders		Taste disturbance	Headache	Dizziness, paraesthesia, somnolence	
Eye disorders				Blurred vision	
Ear and labyrinth disorders				Vertigo	
Respiratory, thoracic and mediastinal disorders		Bronchospasm			
Gastrointestinal disorders		Stomatitis, gastrointestinal candidiasis	Abdominal pain, constipation, diarrhoea, flatulence, nausea/ vomiting, fundic gland polyps (benign)	Dry mouth	Microscopic colitis
Hepatobiliary disorders	Hepatic failure, encephalopathy in patients with pre-existing liver disease	Hepatitis with or without jaundice		Increased liver enzymes	

Category	Classification				
	Very rare	Rare	Common	Uncommon	Not known
Skin and subcutaneous tissue disorders	Erythema multiforme, Stevens-Johnson syndrome, toxic epidermal necrolysis (TEN)		Administration site reactions* Rare: Alopecia, photosensitivity	Dermatitis, pruritus, rash, urticaria	Subacute cutaneous lupus erythematosus (SCLE)
Musculoskeletal, connective tissue and bone disorders	Muscular weakness	Arthralgia, myalgia		Fracture of the hip, wrist or spine	
Renal and urinary disorders	Interstitial nephritis				
Reproductive System and breast disorders	Gynaecomastia				
General Disorders and administration site conditions		malaise, increased sweating			

Irreversible visual impairment has been reported in isolated cases of critically ill patients who have received omeprazole (the racemate) intravenous injection, especially at high doses, but no causal relationship has been established. Other possible side effects include Clostridium difficile associated diarrhoea and Vitamin B12 deficiency (with unknown frequency).

SYMPTOMS AND TREATMENT OF OVERDOSE

There is very limited experience to date with deliberate overdose. The symptoms described in connection with an oral dose of 280 mg were gastrointestinal symptoms and weakness. Single oral doses of 80 mg esomeprazole and intravenous doses of 308 mg esomeprazole over 24 hours were uneventful. No specific antidote is known. Esomeprazole is extensively plasma protein bound and is therefore not readily dialyzable. As in any case of overdose, treatment should be symptomatic and general supportive measures should be utilised.

EFFECT ON ABILITY TO DRIVE AND USE MACHINE

ESOME is not likely to affect the ability to drive or use machines.

INSTRUCTION FOR USE

Intravenous Injection (40 mg) over not less than 3 minutes

The freeze-dried powder should be reconstituted with 5 mL of 0.9% Sodium Chloride Injection, Lactated Ringer's Injection or 5% Dextrose Injection. Withdraw 5 mL of the reconstituted solution and administer as an intravenous injection over no less than 3 minutes. The reconstituted solution should be stored at room temperature up to 30°C (86°F) and administered within 12 hours after reconstitution for 0.9 % Sodium Chloride Injection and Lactated Ringer's Injection and administered within 6 hours after reconstitution for 5% Dextrose Injection. No refrigeration is required.

Intravenous Infusion (40 mg) over 10 to 30 minutes

A solution for intravenous infusion is prepared by first reconstituting the contents of one vial with 5 mL of 0.9% Sodium Chloride Injection, Lactated Ringer's Injection or 5% Dextrose Injection and further diluting the resulting solution to a final volume of 50 mL. The solution (admixture) should be administered as an intravenous infusion over a period of 10 to 30 minutes.

The admixture should be stored at room temperature up to 30°C (86°F) and should be administered within the designated time period as listed in the Table below. No refrigeration is required.

Diluent	Administer within:
0.9 % Sodium Chloride Injection	12 hours
Lactated Ringer's Injection	12 hours
5% Dextrose Injection	6 hours

Esomeprazole sodium for Injection should not be administered concomitantly with any other medications through the same intravenous site and/ or tubing. The intravenous line should always be flushed with either 0.9% Sodium Chloride Injection, Lactated Ringer's Injection or 5% Dextrose Injection both prior to and after administration of Esomeprazole sodium for Injection. Parenteral drug products should be inspected visually for particulate matter and discoloration prior to administration, whenever solution and container permit.

INCOMPATIBILITIES

This medicinal product must not be used with other medicinal products except the compatible diluents as prescribed

STORAGE CONDITION

Store below 30°C.

The reconstituted & diluted solution should be stored at room temperature up to 30°C and administered within 12 hours for 0.9 % Sodium Chloride Injection and Lactated Ringer's Injection and administered within 6 hours for 5% Dextrose Injection.

SHELF LIFE

2 years.

The reconstituted & diluted solution should be stored at room temperature up to 30°C and administered within 12 hours for 0.9 % Sodium Chloride Injection and Lactated Ringer's Injection and administered within 6 hours for 5% Dextrose Injection.

THERAPEUTICS CODE

ATC code : A02BC05

PRESENTATIONS

Esomeprazole Sodium for Injection is supplied as a freeze-dried powder/cake containing 40 mg of Esomeprazole per vial.

PACKAGING

The vial is supplied as 1 vial packed in a mono carton along with leaflet

MANUFACTURED BY:

Gland Pharma Limited
Survey No. 143 to 148,150,151
near Gandimaissamma Mandal Cross roads,
D.P.Pally, Dundigal, Medchal-Malkajgiri,
Hyderabad-500043,Telangana,INDIA

PRODUCT REGISTRATION HOLDER:

UNIMED SDN BHD
No. 53, Jalan Tembaga SD 5/2B,
Bandar Sri Damansara,
52200 Kuala Lumpur, Malaysia.

DATE OF REVISION

OCTOBER 2025